Ruochi Li

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Research Interests: Multimodal RAG | LLM Evaluation | AI Agents

EDUCATION

May 2027 **North Carolina State University, Raleigh, NC**
Ph.D. Candidate in Computer Science GPA: 3.9/4.0

May 2023 **Northeastern University, Boston, MA**
Master of Science in Computer Science GPA: 3.8/4.0

May 2021 **North Carolina State University, Raleigh, NC**
Bachelor of Science in Computer Science GPA: 3.5/4.0

WORK EXPERIENCE

May 2026 – Aug. 2026 **Meituan, Shanghai, China**
AI Agent Engineer Intern
LangGraph, DeepAgents, MCP, RAG

- Built multimodal context-compaction middleware in LangGraph and DeepAgents, combining tool-result offloading with model-aware rolling summarization to support million-token agent sessions; reduced LLM inference cost by **32.4%**.
- Implemented progressive loading and tool gating for **70+ skills**, reducing first-turn prompt tokens by **82%**; preserved skill activation across context compression using a dedicated LangGraph state channel.
- Contributed to the design of user-scoped, two-tier long-term memory with per-run context snapshots, on-demand retrieval, and offline consolidation; helped reduce errors in memory-dependent daily tasks by **75%**.
- Built a fixed LangGraph workflow for natural-language data queries, combining BM25/vector entity linking, structured parameters for platform SQL generation, and deterministic permission resolution; achieved **91%** end-to-end query success.

May 2024 – Aug 2024 **Supermap, Beijing, China**
Research Scientist Intern
Python, PyTorch, VLM, Hyperbolic BERT, VLM Alignment & Fine-Tuning, NER

- Fine-tuned LLaVA on remote-sensing imagery with CLIP and GPT-4 based visual-text alignment, improving visual-to-text mapping accuracy for geospatial multimodal understanding.
- Built a location Named Entity Recognition (NER) system using a hyperbolic BERT model, improving fine-grained tagging of spatial entities in complex geographical data.

Jun. 2022 – Aug. 2022 **Karyopharm Therapeutics Inc. Boston, MA**
Data Scientist Intern
Python, R, SQL, Scikit-learn

- Implemented data visualization strategies using R and Python for clinical study to monitor patients and site utilization of digital medicine tools including applications and wearable devices.
- Extracted myelofibrosis patient data from the EDC system and ensured that quality standards were met for adverse event prediction through data processing.
- Analyzed data from patients with myelofibrosis to predict trends in adverse events, reaching 94% accuracy for the results.

PUBLICATIONS

Conference Papers

[1] *NeurIPS'26* **R. Li**, J. Lin, H. Zhang, H. Chen, J. Ding, E. Gehringer, and Y. Zhang. TopoGraphRAG-Bench: Evaluating Multimodal GraphRAG on Layout-Grounded Evidence Reasoning. *40th Conference on Neural Information Processing Systems (NeurIPS)*, 2026. Accepted. [Code]

[2] *EMNLP'26* H. Zhang, **R. Li**, Y. Zhang, Z. Liang, J. Ding, T. Xiao, and H. Chen. MetaGAI: A Large-Scale and High-Quality Benchmark for Generative AI Model and Data Card Generation. *Findings of the Association for Computational Linguistics: EMNLP*, pp. 34910–34931, 2026. [Code]

[3] *WWW'26* H. Zhang, **R. Li**, Z. Liang, M. Sattari, P. Vo, C. Qu, T. Xiao, J. Ding, Y. Zhang, and H. Chen. AdaQE-CG: Adaptive Query Expansion for Web-Scale Generative AI Model and Data Card Generation. *Proceedings of the ACM Web Conference (WWW)*, pp. 4360–4371, 2026. [Code]

[4] *ICDM'25* **R. Li**, H. Zhang, E. Gehringer, T. Xiao, J. Ding, and H. Chen. Unveiling the Merits and Defects of LLMs in Automatic Review Generation for Scientific Papers. *IEEE International Conference on Data Mining (ICDM)*, pp. 1370–1379, 2025. **Best Poster Award**. [Code]

[5] *FIE'25* **R. Li**, J. Cui, and E. Gehringer. Data-Driven Team Assignment in Software Engineering Education: A GitHub-Informed, Difficulty-Aware Approach. *IEEE Frontiers in Education Conference (FIE)*, pp. 1–9, 2025.

[6] *JCDL'25* H. Zhang, **R. Li**, S. Shrestha, S. H. Mamidala, R. Putta, A. K. Aggarwal, T. Xiao, J. Ding, and H. Chen. ReviewGuard: Enhancing Deficient Peer Review Detection via LLM-Driven Data Augmentation. *ACM/IEEE Joint Conference on Digital Libraries (JCDL)*, pp. 207–216, 2025.

[7] *EDM'25* H. Du, **R. Li**, and E. Gehringer. Objective Metrics for Evaluating Large Language Models Using External Data Sources. *18th International Conference on Educational Data Mining (EDM)*, pp. 489–495, 2025.

[8] *EDM'24* Q. Jia, J. Cui, H. Du, P. Rashid, R. Xi, **R. Li**, and E. Gehringer. LLM-generated Feedback in Real Classes and Beyond: Perspectives from Students and Instructors. *17th International Conference on Educational Data Mining (EDM)*, pp. 862–867, 2024. Poster paper.

[9] *EDM'24* Q. Jia, J. Cui, R. Xi, C. Liu, P. Rashid, **R. Li**, and E. Gehringer. On Assessing the Faithfulness of LLM-generated Feedback on Student Assignments. *17th International Conference on Educational Data Mining (EDM)*, pp. 491–499, 2024. Short paper.

[10] *CSEE&T'24* J. Cui, R. Zhang, F. Zhou, **R. Li**, Y. Song, and E. Gehringer. How Much Effort Do You Need to Expend on a Technical Interview? A Study of LeetCode Problem Solving Statistics. *36th International Conference on Software Engineering Education and Training (CSEE&T)*, pp. 1–10, 2024. **Best Paper Award**.

[11] *SIGCSE'24* J. Cui, R. Zhang, **R. Li**, F. Zhou, Y. Song, and E. Gehringer. How Pre-class Programming Experience Influences Students' Contribution to Their Team Project: A Statistical Study. *55th ACM Technical Symposium on Computer Science Education, Vol. 1 (SIGCSE TS)*, pp. 255–261, 2024.

[12] *ITiCSE'24* J. Cui, R. Zhang, **R. Li**, F. Zhou, Y. Song, and E. Gehringer. A Comparative Analysis of GitHub Contributions Before and After an OSS Based Software Engineering Class. *ACM Conference on Innovation and Technology in Computer Science Education, Vol. 1 (ITiCSE)*, pp. 576–582, 2024.

[13] *FIE'24* J. Cui, R. Zhang, Q. Jia, F. Zhou, **R. Li**, and E. Gehringer. A Statistical Study of Female Students in a Software Engineering Class: Preparedness, Performance, and Contribution. *IEEE Frontiers in Education Conference (FIE)*, pp. 1–9, 2024.

[14] *ITiCSE'23* J. Cui, R. Zhang, **R. Li**, Y. Song, F. Zhou, and E. Gehringer. Correlating Students' Class Performance Based on GitHub Metrics: A Statistical Study. *ACM Conference on Innovation and Technology in Computer Science Education, Vol. 1 (ITiCSE)*, pp. 526–532, 2023.

[15] *FIE'23* J. Cui, F. Zhou, R. Zhang, **R. Li**, C. Liu, and E. Gehringer. Predicting Students' Software Engineering Class Performance with Machine Learning and Pre-Class GitHub Metrics. *IEEE Frontiers in Education Conference (FIE)*, pp. 1–9, 2023.

[16] *ICSE-SEET'22* J. Cui, **R. Li**, K. Lou, C. Liu, Y. Xiao, Q. Jia, E. Gehringer, and R. Zhang. Can Pre-class GitHub Contributions Predict Success by Student Teams?. *ACM/IEEE 44th International Conference on Software Engineering: Software Engineering Education and Training (ICSE-SEET)*, pp. 40–49, 2022.

Journal Articles

[17] *JBI* Y. Zhou, **R. Li**, A. Cleveland, J. Ding, K. Sha, and H. Chen. Quality-aware Multi-source Data Fusion and Enhancement for Medical Concept Normalization Using Large Language Models. *Journal of Biomedical Informatics*, article 105093, 2026. [Code]

[18] *JBI* H. Chen, Y. Zhou, **R. Li**, A. M. Illa, A. Cleveland, and J. Ding. A Comprehensive Survey on Medical Concept Normalization: Datasets, Techniques, Applications, and Future Directions. *Journal of Biomedical Informatics*, article 105005, 2026.

[19] *JBI* H. Chen, **R. Li**, A. Cleveland, and J. Ding. Enhancing Data Quality in Medical Concept Normalization Through Large Language Models. *Journal of Biomedical Informatics*, vol. 165, article 104812, 2025. [Code]

Preprints

[20] *arXiv* H. Zhang, **R. Li**, Y. Zhang, T. Xiao, J. Chen, J. Ding, and H. Chen. Evolving Roles of LLMs in Scientific Innovation: Assistant, Collaborator, Scientist, and Evaluator. *arXiv preprint arXiv:2507.11810*, 2025.

Conference Abstracts and Presentations

[21] *SCC/MLA'24* H. Chen, A. Cleveland, J. Ding, and **R. Li**. Evaluating ChatGPT-based Data Augmentation for Medical Concept Normalization. *South Central Chapter of the Medical Library Association Annual Meeting (SCC/MLA)*, 2024. Conference abstract and oral presentation.

SERVICES

Journal Reviewer

- Computer Standards & Interfaces
- IEEE Transactions on Network Science and Engineering
- Journal of Cloud Computing
- The Electronic Library

Conference Reviewer

- Annual Meeting of the Association for Computational Linguistics (ACL) *2026*
- The ACM Web Conference (WWW) *2026*
- IEEE International Conference on Data Mining (ICDM) *2025*
- ACM/IEEE Joint Conference on Digital Libraries (JCDL) *2025*

Volunteer

- IEEE International Conference on Data Mining (ICDM) *2025*

TEACHING EXPERIENCE *Teaching Assistant*

CSC 216 Software Development Fundamentals

CSC 517 Object-Oriented Design and Development

CSC 422 Automated Learning and Data Analysis

AWARDS

- **Financial Assistance Award** *2026*
  40th Conference on Neural Information Processing Systems (NeurIPS)
- **Best Poster Award** *2025*
  25th IEEE International Conference on Data Mining (ICDM)
- **Student Travel Award** *2025*
  25th IEEE International Conference on Data Mining (ICDM)
- **Best Paper Award** *2024*
  36th International Conference on Software Engineering Education and Training (CSEE&T)
- **Paper Award (Third Prize)** *2024*
  South Central Chapter of the Medical Library Association (SCC/MLA)